

Chapter 110 — Overactive Bladder — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Overactive bladder (OAB)** = urgency $\pm$ frequency/nocturia $\pm$ urgency urinary incontinence (UUI), **no urinary tract infection (UTI) or other pathology**
- **OAB = symptom; detrusor overactivity (DO)** = urodynamic finding
- **Minimum work-up** (American Urological Association/SUFU [AUA/SUFU]): history, examination, urinalysis
- **No urodynamics/cystoscopy/ultrasound** in uncomplicated initial work-up
- **Urodynamics only if result changes invasive/irreversible decision**
- **$\beta 3$ agonist preferred** (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction [SUFU]); antimuscarinic >3 months $\rightarrow$ **10–65% $\uparrow$ dementia**
- **Extended-release** antimuscarinics preferred
- **Posterior tibial nerve stimulation (PTNS)** ~60–80%; **toxin incontinence** $\downarrow$ ~59%; **sacral nerve stimulation (SNS)** 5-year ~70–80%
- **Toxin retention:** ~20% women, **up to 35% men**; each +10 mL post-void residual (PVR) $\rightarrow$ ~27% $\uparrow$ catheterisation risk
- **No treatment is acceptable** after counselling

TERMINOLOGY AND DEFINITIONS

- **Daytime frequency:** 7 voids = upper limit normal; bother ≥ 8
- **Nocturia:** ≥ 1 waking to void; bother ≥ 2 , marked ≥ 3
- **Urgency:** sudden, compelling, **difficult to defer**
- **UUI most bothersome** lower urinary tract symptom (LUTS), both sexes
- **Urge incontinence** $\rightarrow$ renamed urgency incontinence
- **Mixed UI (MUI)** = stress UI (SUI) + UUI; SUI + "OAB dry" $\neq$ MUI
- **Urinary + faecal** = double/dual incontinence
- **OAB fear** = leakage; **bladder pain syndrome (BPS)** fear = pain
- **BPS:** pain rises steadily; **no UUI**

EPIDEMIOLOGY AND ECONOMICS

- **OAB 11.8–28.7%; UUI 1.5–17.3%**
- **NOBLE (National Overactive Bladder Evaluation):** OAB 16.5% (similar by sex); **UUI 7.6% women vs 2.6% men**
- **EpiLUTS:** highest (28.7%); **EPIC (International Continence Society definitions):** lowest (11.8%)
- **Women:** OAB 1.5–2 $\times$, **UUI 2–3 $\times$ men**
- **Age:** women rise 40s, men 50s–60s; **>75 years ~40% men, 1/3 women**
- **Highest prevalence:** Black men and women
- **Incidence 3–8%/year;** dry $\rightarrow$ wet ~1%/year; remission ~6%/year
- **<6% take medication**
- **~92% stop antimuscarinic within 2 years;** half never refill
- **Stopping reasons:** unmet expectations > side effects > cost
- **Most cost-effective third-line:** onabotulinumtoxinA 100 U; SNS costliest
- **US cost ~\$66 billion (2007)**

PATHOPHYSIOLOGY AND AETIOLOGY

- **DO:** only ~half have urgency; **urgency $\rightarrow$ DO in 44–69%**
- **No animal model of OAB**
- **Main regulatory level:** midbrain/brainstem (periaqueductal grey, pontine micturition centre)
- **A δ** = small myelinated, volume sensing; **C** = unmyelinated, "silent", sensitised in OAB
- **Two bladder theories:** urothelium-based vs myogenic (patchy denervation, micromotions)
- **Bowel–bladder cross-talk:** constipation worsens OAB
- **Bladder outlet obstruction (BOO)** not direct cause — similar OAB prevalence in women
- **Anxiety $\rightarrow$ $\uparrow$ temporal summation** (central sensitisation)

CLINICAL ASSESSMENT

- **Antimuscarinic relative contraindications:** uncontrolled closed-angle glaucoma, cognitive impairment, retention, gastric stasis
- **$\beta 3$ relative contraindications:** uncontrolled hypertension, arrhythmia
- **Occult neurology clue:** new OAB + **erectile dysfunction or tremor**
- **Faecal incontinence** ~50% of women with OAB
- **Pelvic/genital $\pm$ digital rectal examination** = most important examination
- **Frailty: Timed Up and Go;** cognition: **Mini-Mental State Examination**
- **PVR not needed** before first-line/oral therapy if uncomplicated; **bladder scanner** when needed
- **Diary 3–7 days;** minimum 24–48 h
- **Max voided volume** = functional capacity
- **International Consultation on Incontinence Questionnaire (ICIQ)** diary sensation: 0 social, 1 normal, 2 urgency passed, 3 urgency reached toilet dry
- **AUA/SUFU questionnaires:** Urogenital Distress Inventory (UDI), UDI-6, Incontinence Impact Questionnaire (IIQ), **OAB-q**
- **ICIQ-OAB 4 items;** **OAB-V8** = screening
- **Lower Urinary Tract Research Network Symptom Index:** both sexes; 29 and 10 items
- **DO correlates best with UUI,** worst with nocturia
- **OAB wet men $\rightarrow$ DO nearly all;** women with UUI: **DO absent 30–40%**
- **Severe low-volume DO** hiding SUI $\rightarrow$ **refill supine**
- **DO after transurethral resection of prostate** up to ~60%

MANAGEMENT

- **Fluid $\downarrow$ 25%** meaningfully helps
- **Caffeine $\downarrow$** helps urgency/frequency, not consistently UI
- **Bariatric surgery:** ~7–8% $\downarrow$ UI risk per 5% weight lost
- **Pelvic floor muscle therapy (PFMT)/bladder training:** UI $\downarrow$ 50–80%
- **SUFU preferred:** $\beta 3$, extended-release trospium, darifenacin, fesoterodine
- **Men with voiding LUTS:** **$\alpha 1$ -blocker first-line;** add antimuscarinic/ $\beta 3$ for storage
- **Combination retention risk up to ~3% $\rightarrow$ caution if PVR raised**
- **PTNS:** weekly $\times$ **12 weeks;** response weeks 8–10
- **Toxin:** UTI ~20%; ~1/4 completely dry
- **SNS revision >30% at 5 years;** battery replacement inevitable
- **Trial duration before judging efficacy:** 4–12 weeks

- **Augmentation/diversion:** last resort, little non-neurogenic data
- **International Consultation on Incontinence (ICI) pathway, men:** treat **major component** (BOO) first
- **ICI pathway women:** treat **most bothersome** symptom first
- **Frail elderly:** active case finding; treat comorbidity; **wet checks/frequency-volume chart if nocturia**

CLASSIC TRAPS

- **Trap:** OAB needs DO on urodynamics — **Actually:** symptom diagnosis
- **Trap:** SUI plus urgency = MUI — **Actually:** MUI needs UUI
- **Trap:** Urinary + faecal = mixed incontinence — **Actually:** double incontinence

- **Trap:** BOO causes OAB, so TURP cures it — **Actually:** DO persists in many
- **Trap:** Urodynamics for every OAB patient — **Actually:** only if decision-changing
- **Trap:** PVR before starting oral drugs — **Actually:** only if complicated
- **Trap:** Absent DO in woman with UUI excludes OAB — **Actually:** absent in 30–40%
- **Trap:** AUA/SUFU lines must be followed in order — **Actually:** not linear; combine
- **Trap:** C fibres normally signal fullness — **Actually:** A δ do; C silent
- **Trap:** Any prevalence comparison valid — **Actually:** definitions differ across studies